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Silkworm Pupae as a Sustainable Protein Source for Poultry, Fish and Pet Feed

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Abstract

Silkworm pupae are a major by-product of the sericulture industry generated after reeling of cocoons. Traditionally regarded as waste or used in limited local applications, pupae are now receiving increasing attention as an alternative protein source for animal feeds. Rising costs of conventional protein ingredients such as soybean meal and fish meal, together with concerns over environmental sustainability, have intensified interest in insect-based feed resources. Silkworm pupae are rich in protein, essential amino acids, and lipids, and can be processed into full-fat, defatted, fermented, or hydrolysed products for use in poultry, aquaculture, and pet nutrition. Numerous studies report that silkworm pupae meal can partially replace conventional protein sources without compromising growth performance, feed efficiency, or product quality when diets are properly balanced. However, challenges remain related to variability in composition, processing methods, storage stability, chitin content, safety, and standardization. This review critically examines the nutritional value, processing technologies, feeding responses in poultry, fish, and companion animals, safety considerations, and economic and sustainability aspects of silkworm pupae utilization. It highlights research gaps and outlines future directions for integrating silkworm pupae into commercial feed systems within a circular bioeconomy framework.

Keywords: silkworm pupae, alternative protein, poultry feed, aquaculture feed, pet food, circular economy

Introduction

The livestock, poultry, aquaculture, and pet food industries depend heavily on protein-rich feed ingredients to sustain animal growth, health, and productivity. Conventional protein sources such as soybean meal and fish meal dominate global feed formulations, yet both face increasing challenges related to price volatility, land and water use, deforestation, and overexploitation of marine resources (FAO, 2013; Henry *et al.*, 2015). These concerns have stimulated the search for alternative, sustainable, and locally available protein resources. Insects have emerged as promising candidates due to their high feed conversion efficiency, ability to utilize organic by-products, and favourable nutrient profiles (Van Huis *et al.*, 2013).

Among insect resources, silkworm pupae (SWP) are particularly attractive because they are already produced in large quantities as a by-product of the silk industry. During silk reeling, pupae are separated from cocoons after stifling to prevent moth emergence. In many sericulture regions, these pupae are discarded, underutilized, or used in small-scale applications, leading to both economic loss and environmental burden (Nagalakshmi *et al.*, 2011). Converting silkworm pupae into animal feed ingredients represents a "waste-to-wealth" strategy that supports circular bioeconomy principles while providing an additional income stream for sericulture communities. This review analyses the potential of silkworm pupae as a sustainable protein source for poultry, fish, and pet feed by synthesizing findings from experimental studies and review literature.

Nutritional Composition and Feed Value

Silkworm pupae are characterized by high crude protein content, moderate to high lipid levels, and the presence of essential amino acids and minerals. The reported protein content of pupae varies widely depending on species, strain, larval diet, and processing methods, typically ranging from 45–60% in full-fat meal and exceeding 60% in defatted products (Nagalakshmi *et al.*, 2011; Ji *et al.*, 2016). The amino acid profile is generally well balanced and includes substantial levels of lysine, leucine, isoleucine, and valine, making pupae meal suitable for monogastric nutrition (Ramos-Elorduy *et al.*, 2002).

The lipid fraction of silkworm pupae contributes considerable metabolizable energy and is often rich in unsaturated fatty acids, including linoleic and α-linolenic acids (Tomotake *et al.*, 2010). These fatty acids can improve dietary energy density and may influence the fatty acid composition of animal products such as poultry meat and fish flesh. However, high levels of unsaturated fat also increase susceptibility to oxidative rancidity during storage (Ji *et al.*, 2016). In addition, silkworm pupae contain chitin, a structural polysaccharide that may reduce protein digestibility at high inclusion levels, although low to moderate concentrations may exert functional effects on gut health and immunity (Makkar *et al.*, 2014).

Processing Technologies and Product Forms

Fresh silkworm pupae deteriorate rapidly due to microbial growth and enzymatic activity; therefore, stabilization and processing are essential before incorporation into feed. Common processing methods include boiling, steaming, roasting, sun drying, and hot-air drying, followed by grinding into meal (Nagalakshmi *et al.*, 2011). Proper heat treatment reduces microbial contamination and deactivates enzymes, while rapid drying minimizes spoilage and mycotoxin development.

Silkworm pupae are utilized in several product forms. Full-fat pupae meal retains natural oil content and provides high energy but is more prone to oxidation during storage. Defatted pupae meal, obtained through mechanical pressing or solvent extraction, offers higher protein concentration and improved shelf life, making it more suitable for precise feed formulation (Ji *et al.*, 2016). Fermented or hydrolysed pupae products are increasingly explored to enhance digestibility, reduce off-odors, and potentially improve palatability and gut health in animals (Zhang *et al.*, 2014). Each processing method influences nutrient availability, functional properties, and economic feasibility, emphasizing the need for standardized protocols.

Utilization in Poultry Nutrition

Silkworm pupae meal has been evaluated as a partial replacement for soybean meal and fish meal in broiler and layer diets. Several feeding trials demonstrate that inclusion of pupae meal at low to moderate levels does not adversely affect growth performance, feed intake, or feed conversion ratio when diets are balanced for essential amino acids (Nagalakshmi *et al.*, 2011; Ji *et al.*, 2016). Defatted pupae meal is generally preferred because of its higher protein concentration and reduced risk of lipid oxidation.

In broilers, dietary incorporation of silkworm pupae has been reported to sustain body weight gain and carcass yield comparable to conventional diets, while sometimes influencing meat quality traits such as fatty acid composition and oxidative stability (Ji *et al.*, 2016). However, excessive inclusion may reduce digestibility due to chitin content or alter fat deposition patterns, underscoring the importance of optimal inclusion rates. In layers, pupae meal can support egg production and quality when properly formulated, although long-term feeding requires strict quality control to prevent rancidity and maintain consistent nutrient supply (Nagalakshmi *et al.*, 2011).

Application in Aquaculture

Aquaculture is one of the most promising sectors for insect-based feeds because many fish species naturally consume insects and because replacing fish meal is a key sustainability objective. Studies on carps, tilapia, and other cultured species show that silkworm pupae meal can partially substitute fish meal without compromising growth, survival, or feed efficiency when amino acid balance is maintained (Zhang *et al.*, 2014; Makkar *et al.*, 2014). Defatted pupae meal improves pellet stability and reduces excess lipid leaching, making it particularly suitable for aquafeeds.

Research on shrimp nutrition indicates that silkworm pupae can replace a portion of fish meal while maintaining growth performance and antioxidant capacity, although very high inclusion levels may affect hepatopancreatic health (Zhang *et al.*, 2014). Fermented and hydrolysed pupae products may further enhance digestibility and immune responses in aquatic species. Nevertheless, species-specific responses and long-term health effects require further investigation to establish safe upper limits of inclusion.

Potential in Pet Food

The pet food industry increasingly adopts insect proteins as novel and sustainable ingredients. Silkworm pupae offer high protein density and functional lipid components, making them a potential ingredient for dog and cat foods. However, the pet food sector imposes stricter requirements regarding ingredient traceability, safety, and palatability. A major concern is allergenicity, as silkworm proteins have been reported to cause allergic reactions in sensitive individuals (Ji *et al.*, 2016). Processing techniques such as hydrolysis may reduce allergenic potential, but this requires validation for companion animals. Additionally, strong odors associated with full-fat pupae may reduce acceptance, suggesting that defatted or fermented products may be more suitable for pet food formulations.

Safety, Storage, and Regulatory Considerations

Ensuring microbiological and chemical safety is essential for the commercial use of silkworm pupae in feed. Inadequate drying or storage under humid conditions can lead to microbial contamination and mycotoxin development (Nagalakshmi *et al.*, 2011). Chemical hazards such as heavy metals or pesticide residues may occur depending on environmental conditions and mulberry cultivation practices, necessitating routine monitoring (Makkar *et al.*, 2014). Oxidative rancidity remains a major quality issue for full-fat products and can be controlled through defatting, antioxidant supplementation, and proper packaging.

Regulatory acceptance of insect-based feed ingredients varies among countries. While many regions permit insect meals in aquaculture and poultry feeds, stricter regulations often apply to pet foods. Establishing standardized quality specifications and safety guidelines is therefore crucial for large-scale adoption.

Economic and Sustainability Implications

From an economic perspective, silkworm pupae represent a low-cost raw material in sericulture regions and can reduce dependence on imported protein sources. Organized collection and processing through cooperatives or cluster-based models can enhance value addition and rural employment (Nagalakshmi *et al.*, 2011). Environmentally, utilizing pupae as feed reduces waste from the silk industry and contributes to resource efficiency and circular economy principles (Van Huis *et al.*, 2013). By partially substituting fish meal and soybean meal, silkworm pupae also help mitigate pressure on marine ecosystems and agricultural land.

Research Gaps and Future Directions

Despite promising results, several knowledge gaps remain. Comprehensive nutrient databases that account for species, strain, and processing variations are needed for precise feed formulation. Long-term feeding trials should evaluate not only growth but also gut health, immunity, product quality, and safety. Shelf-life studies under tropical storage conditions are essential, particularly for full-fat products. Further research on allergenicity and palatability is required for pet food applications. Finally, techno-economic analyses of large-scale processing and supply chain integration will support commercialization.

Conclusion

Silkworm pupae constitute a nutritionally rich and environmentally sustainable alternative protein source for poultry, fish, and potentially pet feeds. Their high protein content, favourable amino acid profile, and valuable lipid fraction make them suitable for partial replacement of conventional feed ingredients when diets are properly formulated. Processing technologies such as defatting, fermentation, and hydrolysis improve shelf life, digestibility, and safety. Although challenges related to variability, oxidation, allergenicity, and regulatory acceptance persist, continued research and standardization can facilitate broader adoption. Integrating silkworm pupae into animal feeding systems not only enhances feed sustainability but also strengthens the sericulture value chain through value addition and waste utilization.

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